

# Vishnu Kadiyala

Ph.D. Candidate, Computer Science · University of Oklahoma

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## Research Focus

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Computer Vision and Deep Learning; Image Classification, Object Detection, and Localization; CNN and Transformer-Based Visual Architectures; Deep Learning for Document Analysis; Scalable ML Pipelines for Image and Video Data.

## Education

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**University of Oklahoma**, Norman, OK Expected May 2027  
Ph.D. in Computer Science

**University of Oklahoma**, Norman, OK May 2022  
M.S. in Electrical and Computer Engineering  
*Thesis: Localization of Tables and Plots in Documents Using Deep Neural Networks*

**KLE Technological University**, India May 2019  
B.E. in Electronics and Communication Engineering

## Technical Expertise

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**Computer Vision:** Image Classification, Object Detection, Localization, CNNs, U-Nets, Vision Transformers (ViT), Image Processing, OpenCV

**Deep Learning Frameworks:** PyTorch, TensorFlow, Keras

**Programming:** Python, MATLAB

**Data & Tools:** Pandas, NumPy, Matplotlib, Git, Linux

**Infrastructure:** High-Performance Computing, SLURM, Reproducible ML Pipelines

## Research Experience

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**Document Understanding with Deep Learning** Aug 2021 – May 2022

- Designed CNN-based neural architectures for **object detection and localization** of tables and plots in documents, achieving **99% detection accuracy**.
- Constructed high-quality annotated datasets with bounding-box labels and automated document generation pipelines for scalable training data.
- Master's Thesis: Localization of Tables and Plots in Documents Using Deep Neural Networks.*

**NSF AI2ES — Computer Vision & ML Researcher** ([GitHub](#)) 2023 – 2025

- Developed a **vision-based atmospheric visibility estimation** system using outdoor camera imagery and deep learning for statewide visual inference beyond sparse sensor coverage.
- Built a Transformer-based architecture with custom spatial/temporal embeddings for multi-modal sensor data, achieving a **13× improvement** over the classical Marshall–Palmer baseline.
- Designed and maintained reproducible ML training and evaluation pipelines on HPC infrastructure for large-scale image and sensor datasets.
- Co-authored peer-reviewed paper: *Estimating Statewide Atmospheric Visibility From Camera Images* (AMS 2025).

**NASA GeoCARB — Deep Learning Researcher** ([GitHub](#)) Jan 2021 – May 2023

- Designed **U-Net (CNN) architectures** for methane hotspot detection from satellite imagery, achieving **95% accuracy**.
- Improved anomaly detection from 80% to **90.2%** using diffusion-based generative models on image data.
- Prepared and analyzed large-scale image datasets from satellite observations, including **data augmentation and quality assurance** for model training and evaluation.

## Selected Publications

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### Theses

- **V. P. Kadiyala.** *Localization of Tables and Plots in Documents Using Deep Neural Networks*. Master's Thesis, University of Oklahoma, 2022.

### Peer-Reviewed Conference Papers

- M. X. Sasser, M. Wilson Reyes, **V. P. Kadiyala**, et al. *Estimating Statewide Atmospheric Visibility From Camera Images*. Proceedings of the 105th AMS Annual Meeting, 2025.
- E. Spicer, S. Crowell, F. Xu, **V. P. Kadiyala**, et al. *Exploring the Influence of Local Urban and Industrial Carbon-Based Pollutant Sources...* Proceedings of the 104th AMS Annual Meeting, 2024.

### Under Review

- **V. P. Kadiyala.** *Implicit Coordination via Attention-Based Latent Belief Updates in Decentralized Partially Observable Multi-Agent Systems*. Under review at ICML 2026.

## Industry & Engineering Experience

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**Test Automation Intern — Robert Bosch Engineering & Business Solutions**     Jan 2019 – May 2019

- Developed hardware-in-the-loop (HIL) test automation pipelines for Engine Control Units (ECUs).
- Automated ECU software validation using ETAS LABCAR across hardware and digital fault layers.